

RODRIGO GOMES DE ARAÚJO

(+351) 914 574 743 | contact@rodrigoaraujo.pt | rodrigoaraujo.pt | [linkedin.com/in/rodrigoaraujo9](https://www.linkedin.com/in/rodrigoaraujo9) | github.com/rodrigoaraujo9

SUMMARY

MSc Computer Science student focused on systems where **correctness and robustness** are key, such as protocols, consensus and infrastructure people actively rely on. Built a **deployable proof-of-work blockchain** in Rust and is currently extending session types to **enforce correct usage of cryptographic resources** at compile-time. Works well **autonomously** and **asynchronously** and emphasizes **testing, documentation, and interfacing** in the development process. Comfortable working remotely and async.

EDUCATION

Master's **Sep 2025 – Present**
Computer Science *University of Porto (FCUP)*

Bachelor's **Sep 2022 – Jul 2025**
Computer Science and Engineering *University of Porto (FCUP)*

RESEARCH

CaST (Cryptographically-aware Session Types) **Jul 2026 – Present**
Master's Thesis *FreeST, Haskell, Type Systems, Applied Cryptography*

- Extending the expressiveness of session types to capture **cryptographic resource usage**, so that its misuse in protocols is detected and rejected at compile-time.
- **Resource tracking composes across nested subprotocols** of arbitrary depth and is preserved under sequential composition and recursion.
- Formalizing the extension and providing a **soundness proof**, integrating it into the FreeST compiler, and applying it to smart contracts and protocol specification.

PROJECTS

PoW Blockchain for Decentralized Auctions | Rust, Tokio, LibP2P, Dalek, Docker

- Extensive **research on the state of the art of Bitcoin**, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing**, **Ed25519 signatures**, and **Merkle Trees** for transaction validation.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **autonomous network behavior simulation** with clients, bootstrap nodes, well behaved nodes, and malicious nodes.
- Developed for the Data Systems Security class and graded 19.1/20.

Real-Time Modular Synthesizer | Rust, Tokio, Rodio, Ratatui, SQLite

- **Real-time audio engine** isolated from the UI and control logic, communicating through **lock-free atomics and message passing**. The audio path never locks, allocates or blocks on I/O.
- **Modular**, leveraging Rust's type system. The audio DSP graph is built at runtime and instantiated per note, so voices run **independently** and the sound can be **reconfigured in real-time** without interruption.

CERTIFICATIONS

2nd Place in IEEE RetroJam 2025 | IEEE UP Student Branch **Oct 2025**

- Built a game from scratch in 48 hours, using Rust and Raylib, learning how to **reduce a problem to its core requirements**.

Academic Merit Awards **2020, 2021, 2022**

Participation in RoboCup 2016 | RoboCup Federation **Jun 2016**

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança **May 2016**

TECHNICAL SKILLS

Programming Languages: Rust, Haskell, C/C++, OCaml, Java, SQL

Frameworks & Tools: Git, UNIX, Tokio, Docker, Plotly, POSIX, STM, gRPC, LibP2P, Dalek, PostgreSQL, Terraform, Raylib and OpenMP

Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)